Nonlinear equations

Ágnes Baran, Csaba Noszály

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Examples:

$$\cos(x) - x = 0$$

or

$$x^5 - 3x^4 + x^3 - 5x^2 + 3 = 0$$

or

$$e^x - 4x^2 = 0$$

or

$$\ln(x) - x + 2 = 0$$

1. Method of bisection

We assume that the function $f : [a, b] \to \mathbb{R}$ is continuous, and $f(a) \cdot f(b) < 0$
In this case the equation

$$f(x) = 0$$

has root(s) in (a, b).

Bisection or midpoint method:

- Evaluate the function at the midpoint $\frac{a+b}{2}$ and examine its sign.
- Use the midpoint to replace the limit at which has the same sign of the function.

The algorithm for finding root with bisection

Pseudocode

A maximum for the number of the iteration steps (maxit) and an accuracy ε are given.

1. assign $k = 1$, $x_0 = a$ and $x_1 = b$
2. assign $x_2 = \frac{x_0 + x_1}{2}$
3. a) if $f(x_2) = 0$, then x_2 is root $\rightarrow$ exit
   b) if $f(x_0) \cdot f(x_2) < 0$, then $x_1 = x_2$
   c) if $f(x_1) \cdot f(x_2) < 0$, then $x_0 = x_2$

   if $|x_1 - x_0| < \varepsilon \rightarrow$ exit (x_2)

   $k := k + 1$

   if $k = maxit \rightarrow$ exit (no root has been found)

   $\rightarrow 2.$

Exercise 1

Write a function that implements the midpoint method! Use the following template:

```
% the function 'midPoint' assumes that f(a)f(b)<0, that is
% 'f' changes its sign over '[a,b]'
% for a given 'f' find its root in [a,b] within 'epsi' error
% 'x': the midpoint of the last interval
function x = midPoint(f, a, b, epsi)
```

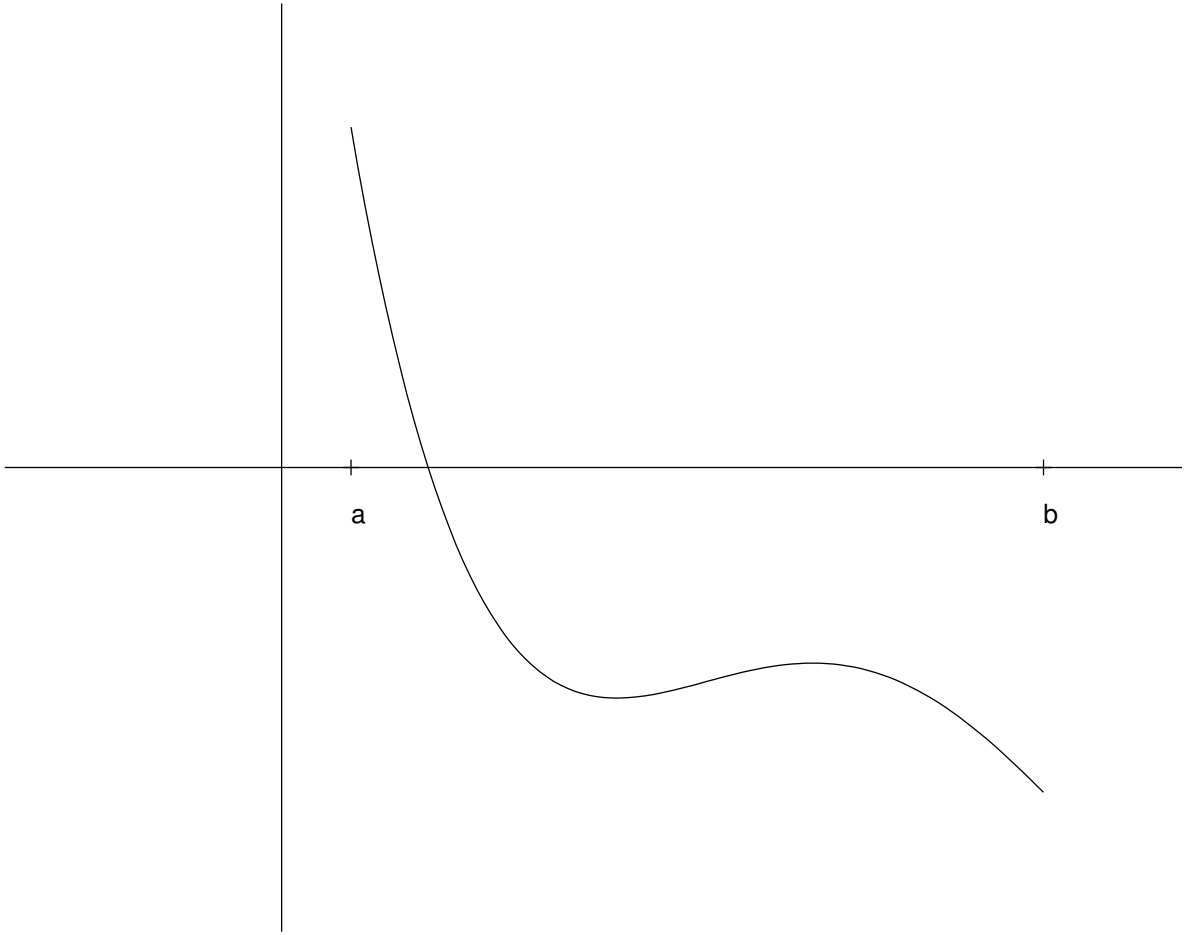
Exercise 2

Find a root for the following functions with the function `midPoint`:

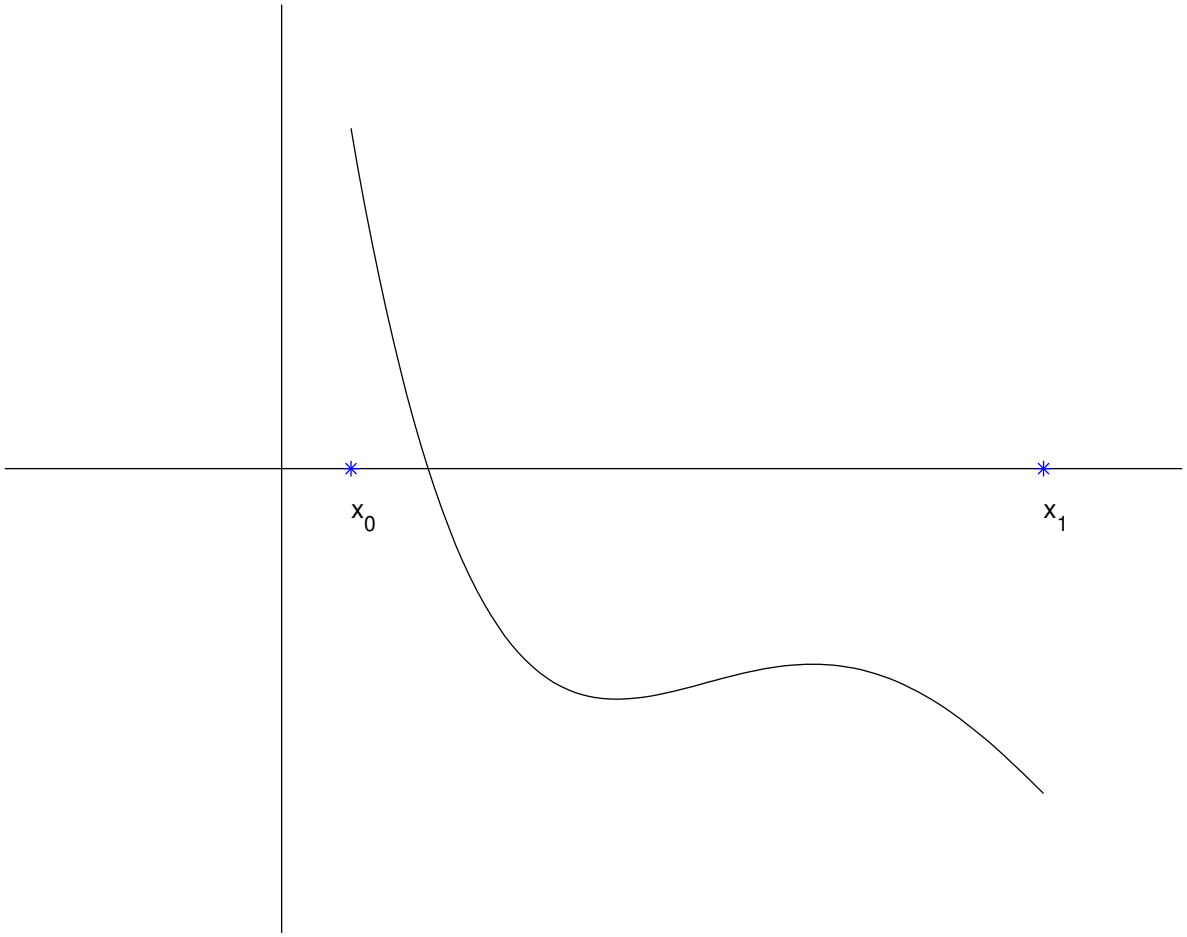
- $cos(x) - x$
- $x^5 - 3x^4 + x^3 - 5x^2 + 3$
- $e^x - 4x^2$
- $\log(x) - x + 2$

2. False position method (regula falsi)

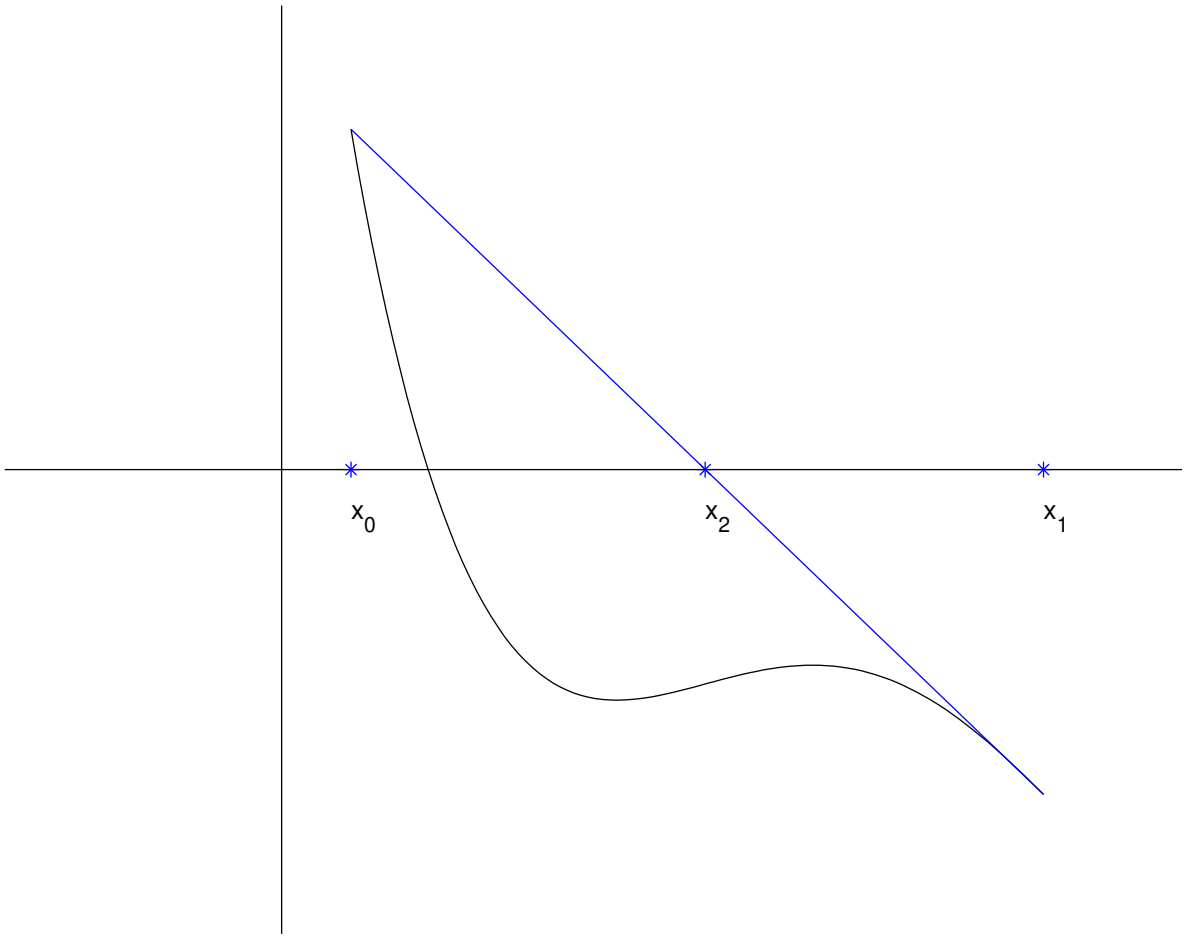
Finds the root of the nonlinear equation $f(x) = 0$, where $f : [a, b] \to \mathbb{R}$, the function f is continuous, and $f(a) \cdot f(b) < 0$.



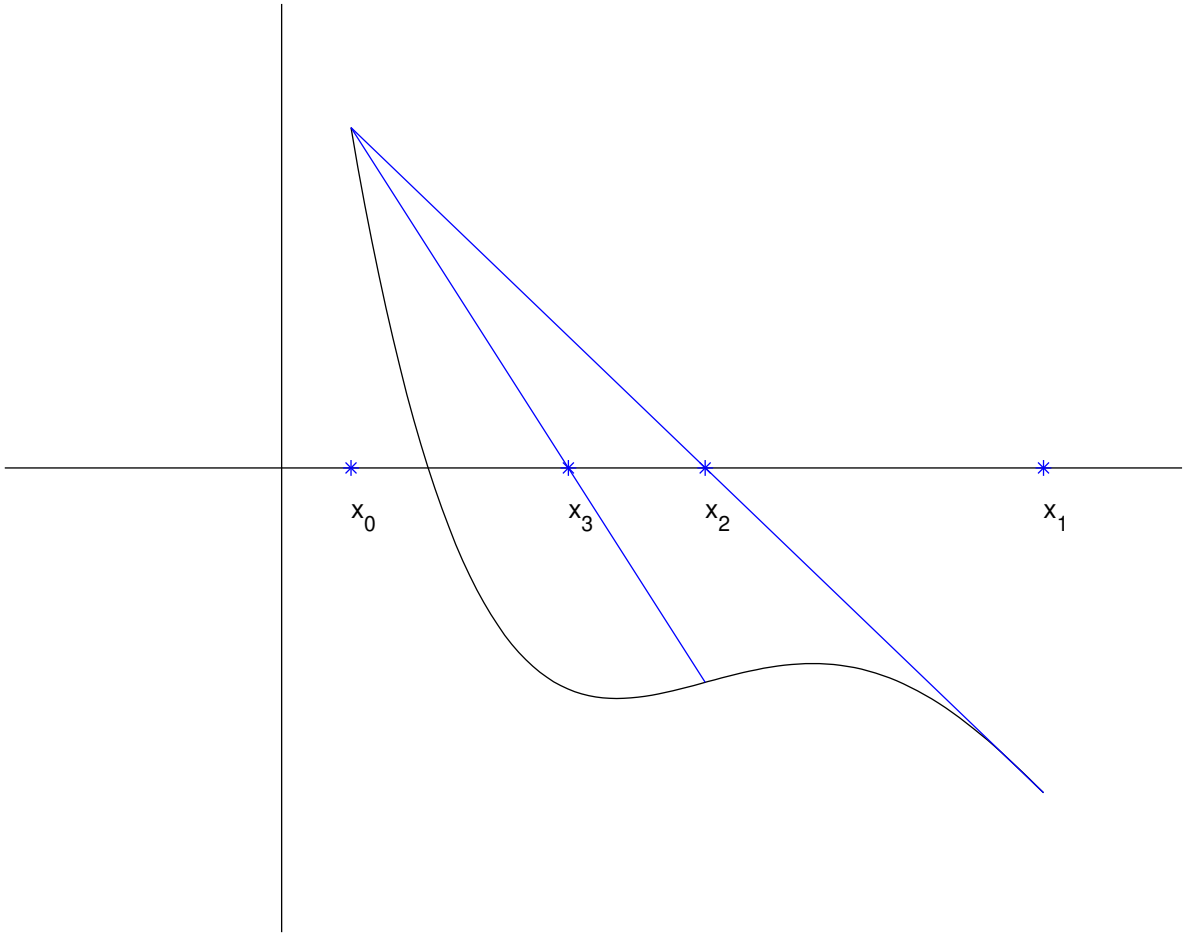
$$x_0 = a \qquad x_1 = b$$

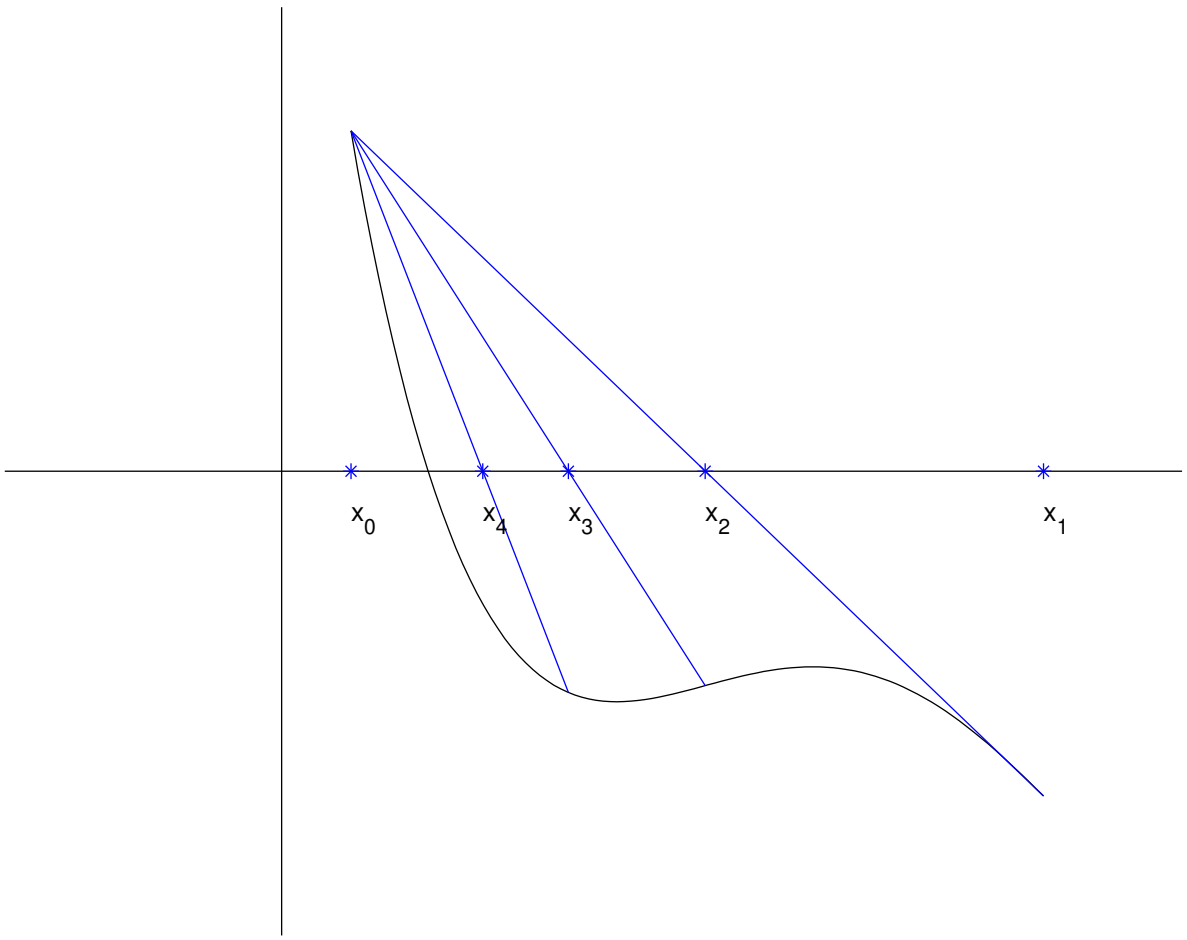


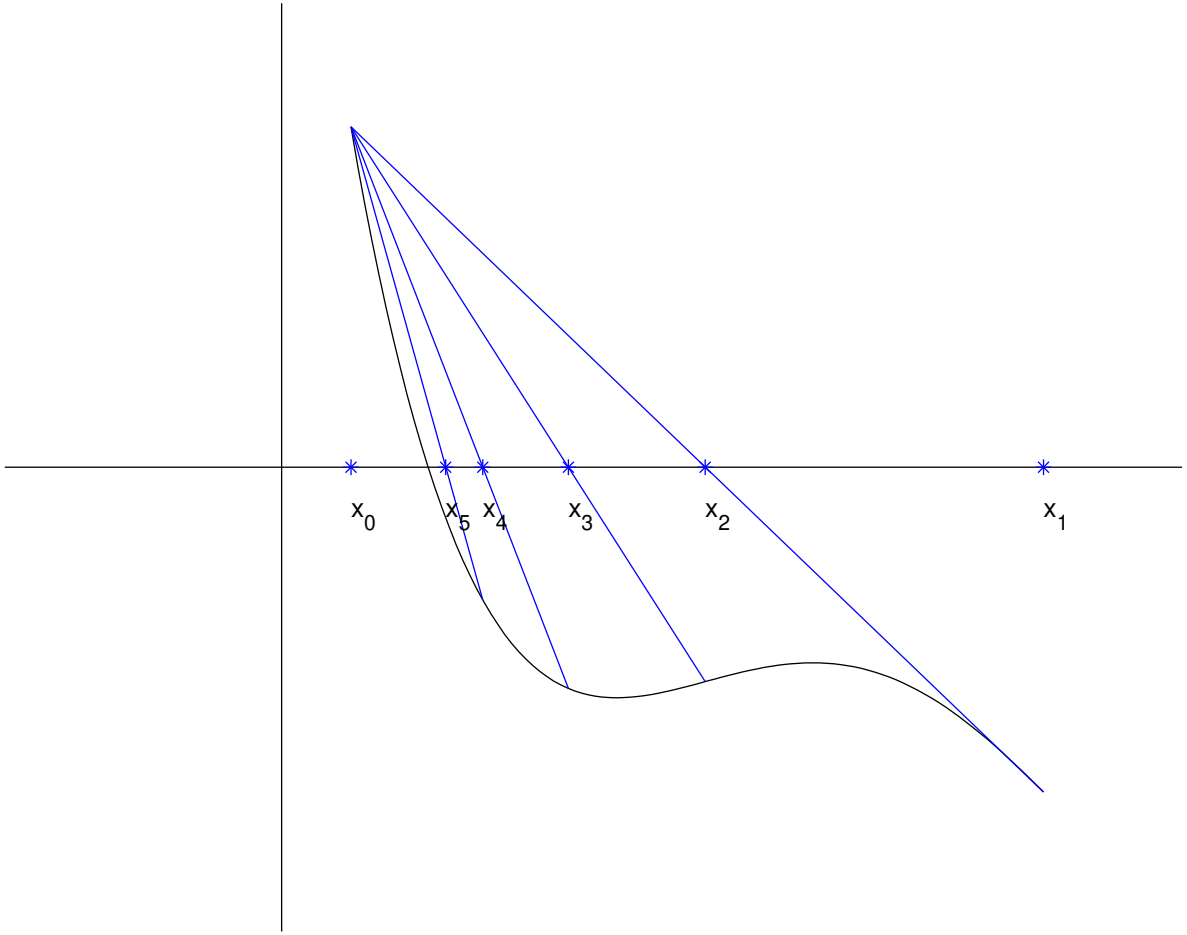
The next position (x_2) is where the secant line crosses the x-axis.



Repeat the algorithm with the subinterval where the sign of function changes.




x_0
x_4
x_3
x_2
x_1


x0
x5
x4
x3
x2
x1

Calculation of the point x_{k+1} when the endpoints of the preceding interval are: x_k and x_j.

We can calculate the line going through the $(x_j, f(x_j))$ and $(x_k, f(x_k))$ points using Lagrange interpolation:

$$\begin{array}{lll} x_j & f(x_j) & \\ & & \frac{f(x_k)-f(x_j)}{x_k-x_j} \\ x_k & f(x_k) & \end{array}$$

$$y(x) = f(x_k) + \frac{f(x_k)-f(x_j)}{x_k-x_j} \cdot (x - x_k)$$

The point x_{k+1} is where $y(x_{k+1}) = 0$, i.e. at:

$$x_{k+1} = x_k - f(x_k) \cdot \frac{x_k - x_j}{f(x_k) - f(x_j)}$$

False position method

Starting points are x_0, x_1. In the next iterations:

$$x_{k+1} = x_k - f(x_k) \cdot \frac{x_k - x_j}{f(x_k) - f(x_j)}, \quad k = 1, 2, \ldots$$

where j is the largest index for $j < k$ for which $f(x_j) \cdot f(x_k) < 0$. x_j and x_k bracket the point where the sign changes.

Properties:
- this formula is well defined,
- for a continuous function f the procedure converges to one of the roots,
- the iteration is based on two points.

Algorithm of the false position method:

Pseudocode

A maximum for the number of the iteration steps (maxit) and an accuracy ε are given.

1. $x_0 := a$, $x_1 := b$, $f0 := |f(x_0)|$
2. $$x_2 := x_1 - f(x_1) \cdot \frac{x_1 - x_0}{f(x_1) - f(x_0)}$$
3. a) If $f(x_2) = 0$, then exit (x_2 is a root).
   b) if $f(x_2) \cdot f(x_1) < 0$, then $x_0 = x_1$, $x_1 = x_2$
   c) if $f(x_2) \cdot f(x_0) < 0$, then $x_1 = x_2$

   if $|f(x_2)| < \varepsilon * (1 + f0)$, then exit ($x_2$ is a root)

   $k := k + 1$

   if $k = maxit$, then exit (no root is found)

   $\rightarrow$ 2.

Exercise 3

Write a function that implements the false position method! Use the following template:

```
% the function 'falsePos' assumes that f(a)f(b)<0, that is
% 'f' changes its sign over '[a,b]'
% for a given 'f' find its root in [a,b] within 'epsi' error
% 'x': the midpoint of the last interval, 'n': number of steps taken
% set 'n' to -1 in case of failure
function [x,n] = falsePos(f, a, b, epsi, maxit)

end
```

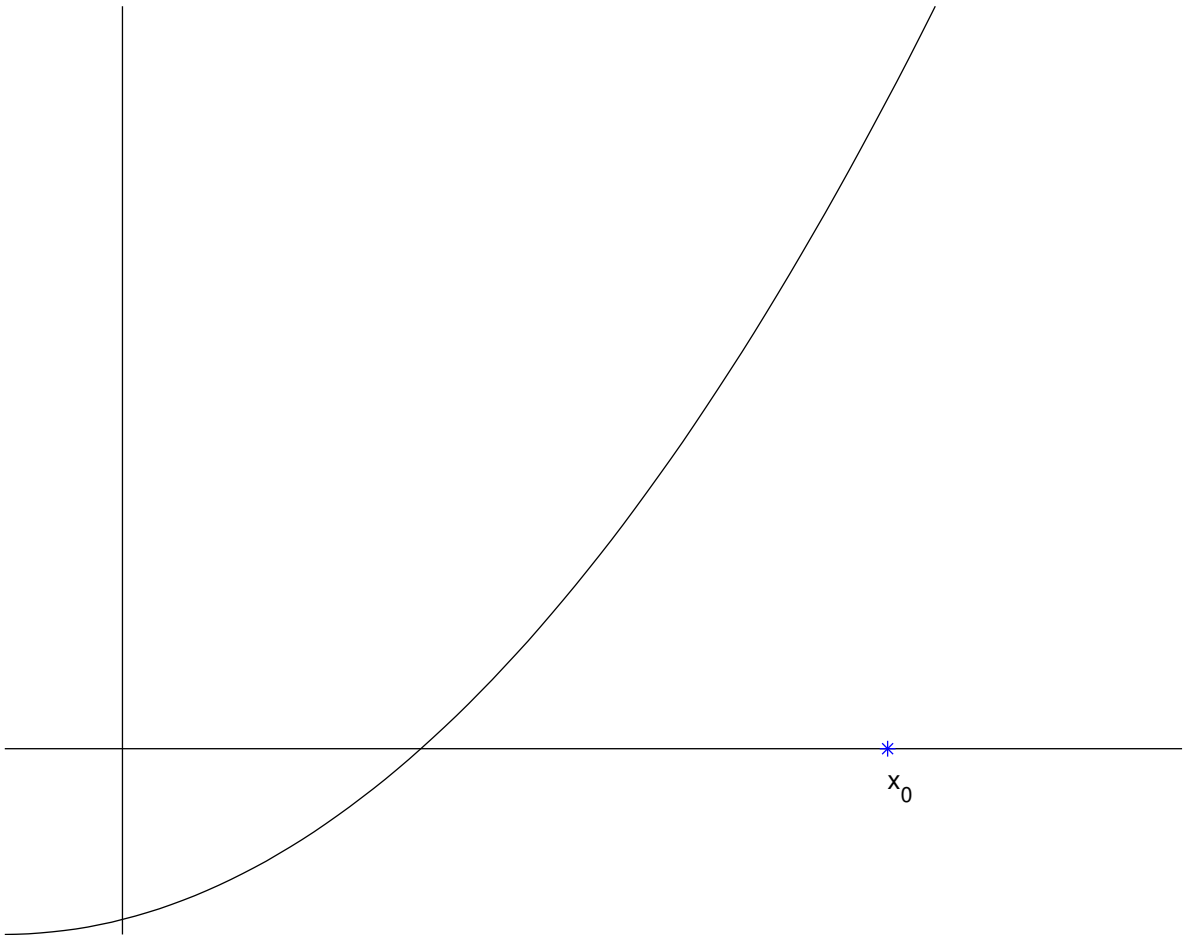
Exercise 4

Find a root for the following functions with the function `falsePos`:

- $cos(x) - x$
- $x^5 - 3x^4 + x^3 - 5x^2 + 3$
- $e^x - 4x^2$
- $\log(x) - x + 2$

3. Newton-Raphson (NR) method

We search for the root of the nonlinear equation $f(x) = 0$.



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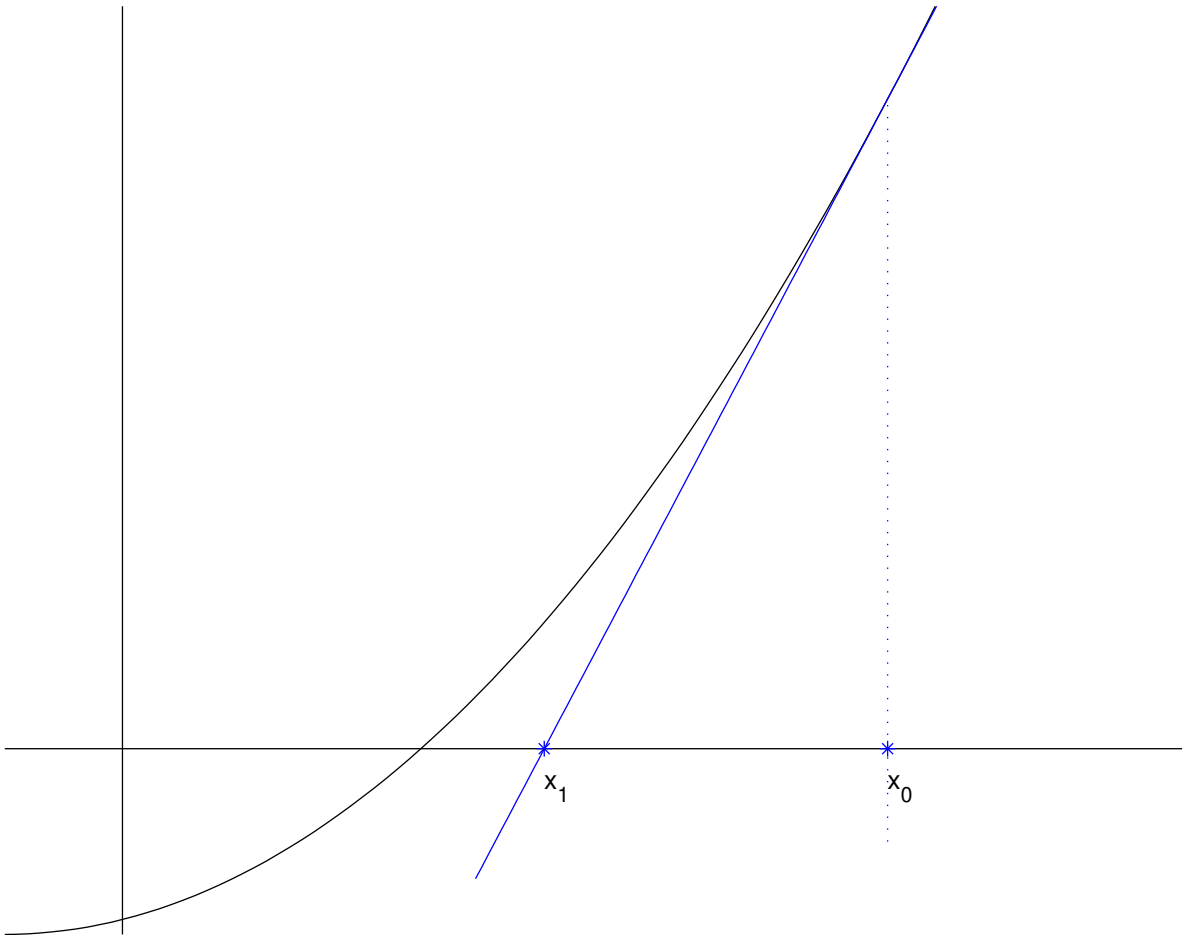

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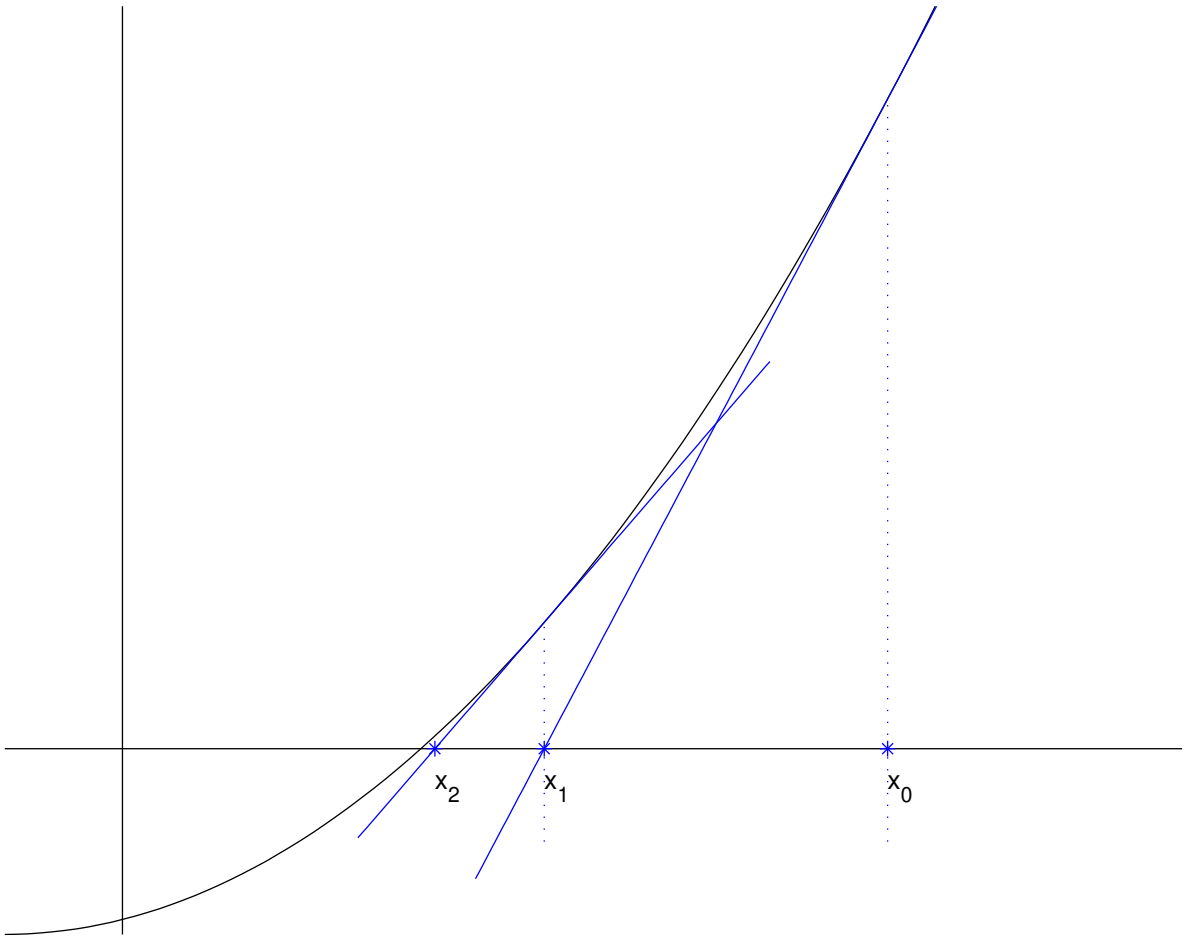

Algorithm of the NR method

Given an initial guess x_0 of the root,

Calculation of the next iteration point x_{k+1} using the tangent of the function f at the point x_k:

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The equation of the straight line going through $(x_k, f(x_k))$ with tangent $f'(x_k)$ is:
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Crossing point of the line with the x-axis is at $y(x_{x+1}) = 0$.
We express x_{k+1} from the condition: $0 = f(x_k) + f'(x_k) \cdot (x_{x+1} - x_k)$.

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} \quad k = 0, 1, \ldots$$

Newton-Raphson method

Given an initial guess x_0

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- NR iteration is based on a single point.
- NR iteration is fast but it might not converge to a root.

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Theorem. Let x^* be a root of f. If

- f is twice continuously differentiable,
- $|f'(x)| \geq m_1 > 0$,
- $|f''(x)| \leq M_2$,
- $|x_0 - x^*| < \frac{2m_1}{M_2}$,

then the NR iteration is well-defined, $x_k \to x^*$, if $k \to \infty$, and

$$|x_{k+1} - x^*| \leq C|x_k - x^*|^2.$$

Example 1

Approximate the value of $\sqrt{a}$, $(a > 0)$ using NR iteration.

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$f(x) = x^2 - a$, and $f'(x) = 2x$ The NR iteration is:

$$x_{k+1} = x_k - \frac{x_k^2 - a}{2x_k} = \frac{1}{2}\left(x_k + \frac{a}{x_k}\right)$$

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Take e.g. $a = 5$, $x_0 = 2$.

$$\begin{aligned} x_1 &= 2.\underline{2}5 \\ x_2 &= 2.\underline{236}11111111111 \\ x_3 &= 2.\underline{23606797}91580 \\ x_4 &= 2.\underline{23606797749979} \end{aligned}$$

Example 2 Approximate the value of the root of the equation $x^3 - 3x - 2 = 0$ using NR iteration starting from the point $x_0 = 1.5$.

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$$x_{k+1} = x_k - \frac{x_k^3 - 3x_k - 2}{3x_k^2 - 3}, \quad k = 0, 1, 2, \ldots$$

$$\begin{aligned}
x_0 &= 1.5 \\
x_1 &= 2.33333333333333 \\
x_2 &= 2.\underline{0}5555555555556 \\
x_3 &= 2.\underline{00}194931773879 \\
x_4 &= 2.\underline{00000}252829797 \\
x_5 &= 2.\underline{00000000000}426
\end{aligned}$$

Example 3 Approximate the value of the root of the equation $x^3 - 4x^2 + 5x - 2 = 0$ using NR iteration starting from the point $x_0 = 2.5$.

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$f(x) = x^3 - 4x^2 + 5x - 2$, and $f'(x) = 3x^2 - 8x + 5$.

Example 3 Approximate the value of the root of the equation $x^3 - 4x^2 + 5x - 2 = 0$ using NR iteration starting from the point $x_0 = 2.5$.

$f(x) = x^3 - 4x^2 + 5x - 2$, and $f'(x) = 3x^2 - 8x + 5$.

$$x_{k+1} = x_k - \frac{x_k^3 - 4x_k^2 + 5x_k - 2}{3x_k^2 - 8x + 5}, \quad k = 0, 1, 2, \ldots$$

$$
\begin{aligned}
x_0 &= 2.5 \\
x_1 &= 2.200000000000000 \\
x_2 &= 2.\underline{0}50000000000000 \\
x_3 &= 2.\underline{00}4347826086957 \\
x_4 &= 2.\underline{0000}37320395597 \\
x_5 &= 2.\underline{00000000}2785312 \\
x_6 &= 2.\underline{000000000000000} \\
x_7 &= 2.\underline{000000000000000}
\end{aligned}
$$

The convergence of the NR iteration is fast to the root $x = 2$.
But it is different if we change the starting point to $x = 1.51$.

$$
\begin{aligned}
x_0 &= 1.51 \\
x_1 &= 0.978297872340423 \\
x_2 &= 0.989034902736187 \\
x_3 &= 0.994487879417566 \\
x_4 &= 0.997236406129453 \\
x_5 &= 0.998616301584373 \\
x_6 &= 0.999307673128352 \\
x_7 &= 0.999653716859631 \\
x_8 &= 0.999826828467474 \\
\hline
x_{22} &= 0.999999980299274
\end{aligned}
$$

The convergence of the NR iteration is poor, because $f(x) = 0$ has a root with multiplicity $t = 2$ at $x = 1$. $f(x) = (x-1)^2(x-2)$.

If we modify the iteration and instead of

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

we use for $t = 2$

$$x_{k+1} = x_k - t\frac{f(x_k)}{f'(x_k)} = x_k - 2\frac{f(x_k)}{f'(x_k)}$$

the convergence becomes faster:

$$
\begin{aligned}
x_0 &= 1.51 \\
x_1 &= 0.446595744680846 \\
x_2 &= 0.916328287619303 \\
x_3 &= 0.996889867448158 \\
x_4 &= 0.999995185996090 \\
x_5 &= 1.000000000044966 \\
x_6 &= 1.000009876002934 \\
x_7 &= 0.999999999978344 \\
x_8 &= 0.999999999978344
\end{aligned}
$$

Exercise 5

Write a function that implements the Newton-Raphson method! Use the following template:

```
% newRap approximates the root of 'f' starting from 'x'
% taking 'maxit' iterations, until an appropriate x found
% 'n': number of steps taken
% set 'n' to -1 in case of failure
% 'df' is the user supplied derivative of 'f'
function [x,n] = newRap(f, df, x, epsi, maxit)

end
```

Exercise 6

Find a root for the following functions with the function `newRap`:

- $cos(x) - x$
- $x^5 - 3x^4 + x^3 - 5x^2 + 3$
- $e^x - 4x^2$
- $\log(x) - x + 2$

4. Secant method

NR method needs the derivative of the function at each step.

$$f'(x_k) \approx \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}$$

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$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} \approx x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$$

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Secant method:

For the solution of $f(x) = 0$.
Given x_0 and x_1

$$x_{k+1} = x_k - f(x_k)\frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}, \quad k = 1, 2, \ldots$$

(The false position method with $j = k - 1$.)

- the secant method is not necessarily well-defined
- the iteration is based on 2 points
- the conditions of the convergence are the same as for the NR iteration, but there is one extra condition: $|x_1 - x^*| < \frac{2m_1}{M_2}$

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The order of the convergence:

$$|x_{k+1} - x^*| \leq C|x_k - x^*|^p,$$

where $p = \frac{1+\sqrt{5}}{2} \approx 1.618$.

($p = 1$ for the false position method and $p = 2$ for the NR method.)

Exercise 5

Write a function that implements the secant method! Use the following template:

```
% newRap approximates the root of 'f' starting from 'x'
% taking 'maxit' iterations, until an appropriate x found
% 'n': number of steps taken
% set 'n' to -1 in case of failure
% 'df' is the user supplied derivative of 'f'
function [x,n] = newRap(f, df, x, epsi, maxit)

end
```

Exercise 6

Find a root for the following functions with the function `secMet`:

- $cos(x) - x$
- $x^5 - 3x^4 + x^3 - 5x^2 + 3$
- $e^x - 4x^2$
- $\log(x) - x + 2$

Example

Built-in functions for root finding:
The function `roots` for finding roots of polynomials:

```
>> p=[1 -2 1 ]
p =

   1 -2 1

>> roots( p )
ans =

   1
   1
```

Example

Built-in functions for root finding:
The function `fsolve` for general one-variable root finding:

```
>> fsolve( @(x) exp (x) -4 *x .^2, 0 )
ans = -0.40778
>> fsolve( @(x) exp (x) -4 *x .^2, 0.5 )
ans = 0.71481
```

You have to supply the initial guess for `fsolve`, it is the second parameter.

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Finds the root of the equation $g(x) = x$, where $g : [a, b] \to \mathbb{R}$.

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The algorithm:
Given the point x_0,

$$x_{k+1} = g(x_k), \quad k = 0, 1, \ldots$$

Theorem. If $g([a, b]) \subseteq [a, b]$, and $\exists \quad 0 \leq q < 1$:

$$|g(x) - g(y)| \leq q \cdot |x - y| \quad \forall x, y \in [a, b], \tag{1}$$

then there exists a unique $x^* \in [a, b]$, that $g(x^*) = x^*$, moreover for all $x_0 \in [a, b]$ the sequence $x_{k+1} = g(x_k)$, $k = 0, 1, \ldots$ converges to x^*.

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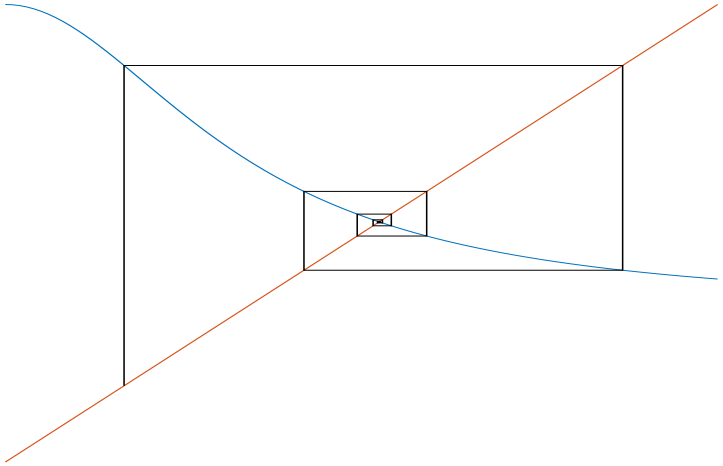
then there exists a unique $x^* \in [a, b]$, that $g(x^*) = x^*$, moreover for all $x_0 \in [a, b]$ the sequence $x_{k+1} = g(x_k)$, $k = 0, 1, \ldots$ converges to x^*.

Remark. If $|g'(x)| \leq q < 1$, then (1) satisfies.



Example. Approximate the root of the equation $\cos x = 3x$ on the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ using fixed point iteration. What can you tell about the convergence of the iteration?

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$$\underbrace{\frac{1}{3}\cos x}_{g(x):=} = x.$$

The iteration: starting from the point x_0

$$x_{k+1} = \frac{1}{3}\cos(x_k), \quad k = 0, 1, \ldots$$

About the convergence:

1. $-\frac{\pi}{2} < -\frac{1}{3} \leq \frac{1}{3}\cos x \leq \frac{1}{3} < \frac{\pi}{2}$, i.e. $g\left(\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]\right) \subseteq \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

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1. $-\frac{\pi}{2} < -\frac{1}{3} \leq \frac{1}{3}\cos x \leq \frac{1}{3} < \frac{\pi}{2}$, i.e. $g\left(\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]\right) \subseteq \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

2. $g'(x) = -\frac{1}{3}\sin x$, i.e.

$$|g'(x)| \leq \frac{1}{3} < 1.$$

About the convergence:

1. $-\frac{\pi}{2} < -\frac{1}{3} \leq \frac{1}{3}\cos x \leq \frac{1}{3} < \frac{\pi}{2}$, i.e. $g\left(\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]\right) \subseteq \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

2. $g'(x) = -\frac{1}{3}\sin x$, i.e.

$$|g'(x)| \leq \frac{1}{3} < 1.$$

The equation has a unique root in the given interval, and for all $x_0 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ the sequence $x_{k+1} = g(x_k)$, $k = 0, 1, \ldots$ converges to this root.

Systems of nonlinear equations (SNLE).

$f(x) = 0$, where $f : \mathbb{R}^n \to \mathbb{R}^n$, i.e.

$$\begin{aligned} f_1(x_1, \ldots, x_n) &= 0 \\ f_2(x_1, \ldots, x_n) &= 0 \\ &\vdots \\ f_n(x_1, \ldots, x_n) &= 0 \end{aligned}$$

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An example with $n = 2$ is: $f : \mathbb{R}^2 \to \mathbb{R}^2$,

$$\begin{aligned} x_1^2 + x_1 x_2 - 3x_2 + 3 &= 0 \\ -3x_1^2 + x_2^2 - x_1 &= 0 \end{aligned}$$

Fixed point iteration for systems

We look for the root of the equation $g(x) = x$, where $g : T \to \mathbb{R}^n$, $T \subseteq \mathbb{R}^n$.

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We look for the root of the equation $g(x) = x$, where $g : T \to \mathbb{R}^n$, $T \subseteq \mathbb{R}^n$.

Example: Let $T = [-\pi, \pi] \times [-\pi, \pi]$ and consider the system

$$\frac{1}{4}\cos(2x_1 - x_2) - \frac{3}{4} = x_1$$
$$\frac{1}{3}\sin(x_1) - \frac{2}{3} = x_2$$

Here $g : T \to \mathbb{R}^2$:

$$g(x_1, x_2) = \begin{pmatrix} g_1(x_1, x_2) \\ g_2(x_1, x_2) \end{pmatrix} = \begin{pmatrix} \frac{1}{4}\cos(2x_1 - x_2) - \frac{3}{4} \\ \frac{1}{3}\sin x_1 - \frac{2}{3} \end{pmatrix}.$$

The fixed point iteration:
Given an initial vector $x^{(0)}$,

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Theorem. If T is a convex domain, $g(T) \subseteq T$, and g is differentiable, moreover $\|J(x)\| \leq q < 1$ for all $x \in T$, then there exists a unique $x^\star \in T$ for which $g(x^\star) = x^\star$ holds and for all $x^{(0)} \in T$ the sequence $x^{(k+1)} = g(x^{(k)})$, $k = 0, 1, \ldots$ converges to $x^\star$.

Example

Consider the system

$$\begin{aligned} -4x_1 + \cos(2x_1 - x_2) &= 3 \\ -3x_2 + \sin x_1 &= 2 \qquad x_1, x_2 \in [-\pi, \pi] \end{aligned}$$

Examine the solvability of the system and the convergence of the iteration

$$\begin{aligned} x_1^{(k+1)} &= -\frac{3}{4} + \frac{1}{4}\cos(2x_1^{(k)} - x_2^{(k)}) \\ x_2^{(k+1)} &= -\frac{2}{3} + \frac{1}{3}\sin x_1^{(k)} \end{aligned}$$

$(k = 0, 1, 2, \dots)$.

$$
\begin{aligned}
x_1 &= \frac{1}{4}\cos(2x_1 - x_2) - \frac{3}{4} \\
x_2 &= \frac{1}{3}\sin x_1 - \frac{2}{3}
\end{aligned}
$$

Here $g : [-\pi, \pi]^2 \to \mathbb{R}^2$,

$$
g(x_1, x_2) = \begin{pmatrix} g_1(x_1, x_2) \\ g_2(x_1, x_2) \end{pmatrix} = \begin{pmatrix} \frac{1}{4}\cos(2x_1 - x_2) - \frac{3}{4} \\ \frac{1}{3}\sin x_1 - \frac{2}{3} \end{pmatrix}.
$$

1. $g([-\pi,\pi]^2) \subseteq [-\pi,\pi]^2$, because

$$\begin{aligned} -\pi &< -1 \le g_1(x_1,x_2) \le -\tfrac{1}{2} < \pi \\ -\pi &< -1 \le g_2(x_1,x_2) \le -\tfrac{1}{3} < \pi. \end{aligned}$$

2. The Jacobian matrix of g:

$$J(x) = \begin{pmatrix} -\frac{1}{2}\sin(2x_1 - x_2) & \frac{1}{4}\sin(2x_1 - x_2) \\ \frac{1}{3}\cos x_1 & 0 \end{pmatrix},$$

$\|J(x)\|_\infty \le \frac{3}{4}$, i.e. the system has a unique root in $[-\pi,\pi]^2$, and the iteration is convergent for all $x^{(0)} \in [-\pi,\pi]^2$.